



Figure 04xv: Evidence (conserved) of conflagration of Nalanda

2a.ii. Site Planning and layout

(Refer to Appendix 02: Documentation of planning and architecture evident in archaeological remains of Nalanda Mahavihara)

A. Descriptions of Excavated Remains

Nalanda Mahavihara was set in a wooded landscape and prior to its destruction covered an area not less than 2 km square⁵. While the exact extent and features like the gates, boundary wall or its libraries (Ratnasagara, Ratnadodhi and Ratnaranjak) remain debatable in the lack of physical evidence⁶, the Archaeological Survey of India has unearthed its most significant core since 1915. The property consists of viharas, chaityas, stupas and shrines set in a primarily agricultural landscape, reminiscent of the original setting⁷.

The excavated area is the nuclei of Nalanda Mahavihara with structures dateable⁸ between 3rd century BCE to 13th century CE and which were elaborately described by its resident scholars like Xuanzang. While invasion and arson have destroyed a major part of the physical infrastructure and delicate details like stucco elements, what remains is a comprehensive physical manifestation of the spectrum of changes in art, architecture, site-layout and rites and rituals associated with Buddhism. Today, what remains is the principle stupa encased in a chaitya (*stupa-chaitya*), four *chaityas*, eleven *viharas* and innumerable votive stupas and shrines which are dateable to Late-Gupta/pre-Pala, Early and Late Pala period. Fragments of stucco-art can be seen on the eastern facade of the Site no. 03 and in the sanctum of each chaitya in the site.

⁵ Dharmakirti's description of Nalanda (in 1235 CE) remains the last eye-witness account of the Mahavihara. He noted that the 84 viharas and 14 chaityas of Nalanda were set ablaze by the army of Bakhtiyar Khilji, reducing the site to ruins (Phuoc, Buddhist Architecture, 2010). While a number of mounds, aligned in a particular pattern have been noted to be distributed in an area of about 2km, suggesting that the Nalanda could have had atleast two more columns of viharas and *chaityas*. Even (Al)though the exact extent is unknown, the area bound by uniformly shaped water-bodies is replete with antiquities and possible sub-surface remains (Kamini, Kulkarni, Raghavaswamy, & Roy, 2007)(Archaeological Survey of India, 1916-2001).

⁶ Only a fraction of what was possibly Nalanda Mahavihara was recovered from under 8 feet of earth (at greatest depth). The land, from under which the site was excavated, was used for cultivation for atleast 500 years before its archaeological potential was identified. This has caused further damage to the fragmented remains and they have either perished or disturbed from their original location. (This has caused much of the remains already in a fragmented state, have perished or dispersed to distant location over time.)

⁷ Nalanda was chronicled as having been (to be) set in an *amra vana* or *pavarika vana* (mango orchard/forest)

⁸ Includes those constructed and or repaired.

verandahs of the upper floor rested. Its surface is brick-paved, divided into quarters and had a well, a brick-lined hearth and a shrine located to its south.

The structure was excavated in a fragile state and was consolidated to ensure physical integrity. Excavation also unearthed a well from the foundation of the northern wall.

d. Site no. 01 (vihara/ residential-cum-scholastic facility)

This vihara was excavated in a fragile state which was consolidated to ensure physical integrity. The northern, southern and western walls were reinforced, details like niches and openings were retained and to stabilise the superstructure, reinforced concrete lintels were inserted as the original wooden ones had perished. The process of excavation revealed as many as 9²⁹ (6th - 12th century CE) levels of occupations i.e. from early Late-Gupta period to Late Pala period and stratigraphic analysis shows that it was inhabited for the longest tenure. The layers of ash, pottery shards, metal figures, heavy brick debris, ashes and finally natural earth corroborate its continuous use, its destruction by fire³⁰ and finally its abandonment. Continuous use is also evidenced by successive expansion and vertical expansion existing wall³¹ by consolidating previous layers. Furthermore inscriptions found from the 8th phase of occupation indicate it to be contemporary to 5th century CE and the top-most layer to the reign of Devapala (9th century CE).

The vihara is entered from a centrally located gateway on the western wall which was bound on the north and south by walls having recessed shrines. The inner wall of the entrance was partly removed to access the lower level of the western verandah and partly retained to support the superstructure. The study of this wall revealed the earliest levels of occupation.

The gateway leads onto the continuous verandah which encloses a quadrangular courtyard. At its south-western corner are the remains of a brick staircase dateable to

²⁹ A Trial Pit was excavated to a depth of 14' below ground level showed 9 levels of occupation. Also 3 layers of habitation levels were identified where each level is demarcated by a drain which channelized water away from the interior when the level was inhabited. The southern wall was also found to extend to a depth of 8 feet below the floor level of the verandah.

³⁰ The destruction of this structure by fire is evidenced by the remains of vitrified concrete debris and fused bricks.

³¹ For example, parts of the shattered walls have been removed to consolidate the base for one above.

g. Supports for scholarship

i. Allocation of space

Apart from enabling scholars to learn from the most learned persons of a discipline, Nalanda provided high quality of infrastructure which further supported their educational endeavors. Recorded in chronicles of Yi Ying, there was a democratic method for distribution of rooms to the scholars by the high priests. The best spaces, considered the rooms at the four corners of the enclosure, were allocated to the scholars who demonstrated their distinction in academics.

At the onset of monsoon an assembly of priests, possibly the Karmadana together which the Acharyas, Adhyakshas and the Kulapati himself convened a session to decide the distribution of rooms. While each scholar was provided with a room, preferences were given to Sthaviras and the other students followed according to their ranks. Scholars were allowed to retain their rooms for a period of one year which and the process was repeated yearly. This not only ensured maintenance of focus among scholars to out-perform themselves but also enabled a sense of detachment (with worldly possessions) to prevail (Mookerji R. K., 1974).

ii. Access to Library

While no archaeological evidence of the library at Nalanda remains, records mention the built ensemble constituting of the Ratnaranjak, Ratnasagara and Ratnadodhi where the latter was noted as a 9 storied structure. These were known to have housed innumerable manuscripts covering a wide range of subjects with special focus on Buddhism. These manuscripts were either acquired in the form of a gift or earned through winning debate convention or collected from a scholar's private collection upon his death.

The library was a melting pot of culture and its stock was enriched by gifts (of manuscripts and texts) but also by the tradition of translation. According to Chinese, Korean and Tibetan scholars, one of the primary reasons for foreign students to learn Sanskrit was to be able to translate manuscripts and carry the wisdom back to their

Comparable organizational system	Comparable result of scholarship (social mobility and continued influence)
Attributes of the aforementioned organization can be summarized as follows: • Autonomy in administrative and economic decision-making • Distinct legal autonomy and status • Curriculum and a system of awarding degrees • Academic specialization in disciplines relevant to its time • Freedom to engage in all existing forms of knowledge⁴ • Strict criteria for admittance⁵ • Emphasis on intellectual capacity and cultivation of knowledge through critical thinking⁶ • Association with the establishment imparted mobility and acceptance in the highest levels of society • Borne out of interaction between 3 groups - scholars who followed the teaching of the Buddha, scholars of other schools of thought and scholars with the context Also refer to (Chapter 02, Section 2av, Parts A and B)	Scholars upon graduating from Nalanda Mahavihara, University of Bologna and Paris were considered prized human resources of their time. They were socially mobile, invited over a large geo-political area to hold the highest posts as administrators, spiritual leaders and educators. The multi-faceted role played by scholars of Nalanda, Bologna and Paris shaped contemporary society and stimulated the culture of establishing pedagogical infrastructure in subsequent years. The success of the Nalanda Mahavihara, University of Bologna and Paris is further reinforced by the fact that their contemporary and those established later either borrowed from it or aspired to reach it or out-shine it. The extent of continued influence of Nalanda Mahavihara and University of Bologna and Paris is another commonality between the three. Comparable social mobility of the graduates and continued influence of these establishments justify the selection for Comparative Analysis. Also refer to (Chapter 02, Part 2av, Section A, Sub-section iii. Studentship and their contributions, Table no. 05)

Available records on Nalanda's organizational system

It also must be noted that Nalanda Mahaviharas existed at a point in time (6th century BCE to 13th century CE) when consistent records for historical documentation were sparse and the site was ravaged, then set ablaze multiple times⁷. It is possible that original records which could give consistent historical narrative of the Mahavihara are lost to time. What is known about the Mahavihara, is based on correlating and rationalizing information from literary sources mainly records of Chinese and Tibetan scholars, archaeological evidences and objects. Of the records, those of Hui Li, Xuanxang Yijing and Dharmapala are most informative and the process of correlation has been explained in Chapter 02, Part 2ai. *Nalanda Mahavihara from literary accounts and Part 2.a.v. Education and Internal Management at Nalanda, part B - Internal Administrative System*. The functioning of Nalanda is also known from information accrued from seals, copper plates and inscriptions. These pieced together indicate the various systems of administration and management of educational, economic and social facets of the Mahavihara much like a corporate structure (Misra, Nalanda (felicitation Volume-III), 2008) (Mehta, 2013).

⁴ The inclusion of all contemporary types of knowledge sets Nalanda apart from other similar establishments (borne out of a specific School of Thought). The emphasis was knowledge and the inclusion and acknowledgement of existing forms of knowledge was important to participate in tarka or debates. Tarka of debate is a dialogue based learning system where each party must internalize both perspectives completely. Scholars of Nalanda were known to excel in such debates and developed it as a tool for critical thinking (Refer to Chapter 02, part 2av.c. sub-part I. Mode of teaching).

⁵ It is important to note that mastering of all existing forms of knowledge was a prerequisite to gain admittance at Nalanda Mahavihara. Therefore, education here always started after completing the *ashrama* or an equivalent phase of education where the youth would be taught by a number of subject specialists. The latter is similar to the systems and scope of education at cathedral-schools or monastic complexes as here the students would be taught by fathers and nuns subjects approved by institution.

⁶ Debate as a tool for critical thinking was excelled and developed further in Nalanda. This is further explained in Chapter 02 Section 2av., Part A, sub-part 3i.

⁷ A substantial amount of ash was noted during excavation at Nalanda (Archaeological Survey of India, 1916-2001). According to its scholars, at Nalanda Mahavihara were the Ratnasagara, Ratnaranjak and Ratnadodhi, its library. These were replete with innumerable manuscripts that were set ablaze. It is possible that much of the records that could further substantiate our understanding did not survive the ravages of time and man.